

**CO5420: Artificial Neural Networks and Deep
Learning**

**Project Proposal– Human Activity Recognition
Using Smartphone Sensor Data**

GROUP 5

Group Details

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Motivation / Problem Definition

Human Activity Recognition (HAR) is an important application of machine learning and deep learning that enables systems to automatically identify physical activities performed by users based on sensor data. Modern smartphones are equipped with accelerometers and gyroscopes that continuously capture motion-related information, making them ideal platforms for activity recognition.

Accurate activity recognition has numerous real-world applications including fitness tracking, healthcare monitoring, elderly care, rehabilitation systems, smart homes, and context-aware mobile applications. Traditional machine learning approaches rely heavily on manually engineered features extracted from sensor signals. Recent advances in deep learning allow models to learn meaningful representations directly from data, potentially improving recognition accuracy and reducing the need for manual feature engineering.

This project aims to investigate and compare traditional machine learning techniques and neural network-based approaches for human activity recognition using smartphone sensor data. Furthermore, we will evaluate whether end-to-end learning from raw sensor signals provides advantages over using handcrafted features.

Dataset Description

The project will use the **Human Activity Recognition Using Smartphones Dataset** from the UCI Machine Learning Repository.

Dataset Characteristics

- Total instances: 10,299
- Number of activities: 6
- Number of engineered features: 561
- Sensors used:
 - Accelerometer
 - Gyroscope
- Sampling frequency: 50 Hz

Activity Classes

1. Walking
2. Walking Upstairs
3. Walking Downstairs
4. Sitting
5. Standing
6. Laying

Available Data

The dataset contains:

- Preprocessed feature vectors consisting of 561 statistical and frequency-domain features.
- Activity labels corresponding to each observation.
- Raw accelerometer and gyroscope signals for sequence-based modelling.

Dataset Suitability

This dataset is widely used as a benchmark for HAR research and provides both handcrafted features and raw sensor measurements, allowing direct comparison between traditional machine learning methods and deep learning approaches.

Proposed Approach (Initial Ideas)

This project aims to investigate whether deep learning models can outperform traditional machine learning approaches for Human Activity Recognition using smartphone sensor data.

Stage 1 – Baseline Modelling

Traditional machine learning models including Decision Tree and Support Vector Machine (SVM) will be trained and evaluated using the provided 561 engineered features. These models will establish baseline performance.

Stage 2 – Deep Learning with Engineered Features

A Feedforward Neural Network (FNN) will be developed using the same feature representation. The objective is to evaluate whether neural networks can improve recognition performance over traditional machine learning methods.

Stage 3 – End-to-End Sequence Learning

The project will then use raw accelerometer and gyroscope signals to train an LSTM model. This stage investigates whether direct learning from sequential sensor data provides advantages compared with handcrafted feature extraction.

Stage 4 – Sensor Contribution Analysis

Experiments will be conducted using accelerometer-only data, gyroscope-only data, and combined sensor inputs to analyse the contribution of each sensor to activity recognition performance.

Evaluation Metrics and Baseline

All approaches will be evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix analysis. Final comparison will identify the most effective approach for Human Activity Recognition.

Proposed Contribution (Division of Work)

Member 1 (E/23/352) – Dataset Preparation & Preprocessing

- Load and explore the dataset
- Perform data cleaning and preprocessing
- Apply feature normalization and scaling
- Prepare train, validation, and test datasets

Member 2 (E/23/384) – Baseline Machine Learning Models

- Implement Decision Tree classifier
- Implement Support Vector Machine (SVM)
- Perform model training and initial evaluation
- Compare baseline model performance

Member 3 (E/23/427) – Feedforward Neural Network (FNN)

- Design and implement the FNN architecture
- Train the neural network using engineered features
- Perform hyperparameter tuning
- Analyse training and validation performance

Member 4 (E/23/211) – LSTM-Based Activity Recognition

- Process raw accelerometer and gyroscope signals
- Design and implement the LSTM model
- Train and evaluate sequence-based models
- Compare LSTM performance with FNN and baseline models

Member 5 (E/23/435) – Evaluation, Kaggle Submission & Documentation

- Calculate evaluation metrics (Accuracy, Precision, Recall, and F1-score)
- Generate confusion matrices and result visualizations
- Prepare Kaggle submissions and leaderboard analysis
- Compile project report, documentation, and presentation materials

Generative AI Usage Statement

Generative AI tools (ChatGPT) were used during the preparation of this project proposal to assist with idea generation, project planning, task organization, and language refinement. AI-generated suggestions were critically reviewed, modified, and validated by the group members before inclusion in the proposal. All technical decisions, methodology selection, dataset analysis, and project planning reflect the group's own understanding and judgment.

The group acknowledges the module policy regarding the use of generative AI and confirms that AI was used only as a supporting tool for brainstorming, drafting, and refinement, while the final content represents the group's own work.

References

1. I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.
2. UCI Machine Learning Repository, "Human Activity Recognition Using Smartphones Data Set." Available:
<https://archive.ics.uci.edu/ml/datasets/human+activity+recognition+using+smartphones>
3. C. Cortes and V. Vapnik, "Support-Vector Networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.